

Exercise - 06

Aim : To write a program to implement error detection and correction using HAMMING code concept. Make a test run to input data stream and verify error correction feature.

code :

```
def hamming_encode(data):
```

```
    d1, d2, d3, d4 = map(int, data)
```

```
    p1 = d1 ^ d2 ^ d4
```

```
    p2 = d1 ^ d3 ^ d4
```

```
    p4 = d2 ^ d3 ^ d4
```

```
    code = f"{p1}{p2}{d1}{p4}{d2}{d3}{d4}"
```

```
    return code
```

```
def hamming_decode(code):
```

```
    bits = list(map(int, code))
```

```
    s1 = bits[0] ^ bits[2] ^ bits[4] ^ bits[6]
```

```
    s2 = bits[1] ^ bits[2] ^ bits[5] ^ bits[6]
```

```
    s4 = bits[3] ^ bits[4] ^ bits[5] ^ bits[6]
```

```
    error_position = s4 * 1 + s2 * 2 + s1
```

```
    if error_position != 0:
```

```
        print("Error detected at position :", error_position)
```

```
        bits[error_position - 1] ^= 1
```

```
        print("corrected code: ", " ".join(map(str, bits)))
```

else :

print ("No error detected.")

data = f"{bits[2]}{bits[3]}{bits[4]}{bits[5]}{bits[6]}{bits[7]}"

return data

data = input ("Enter 4-bit data : ")

code = hamming_encode (data)

print ("Hamming code : ", code)

received = input ("Enter received 7-bit code : ")

corrected_data = hamming_decode (received)

print ("Original data : ", corrected_data)

Input :

1011

Output :

Enter 4-bit data : 1011

Hamming code : 0110011

Enter received 7 bit code : 0110011

No error detected

Original data : 1011

Part 2

Result : waiting and implementing error detection and correction using hamming code is done successfully.